

Exercises

1. Find mean, median, mode and range

Arr: 12, 15, 18, 15, 20, 22, 15, 25

$$\text{Mean} = \frac{142}{8} = 17.75$$

$$\text{Median} = \frac{(15+18)}{2} = 16.5$$

Mode = most repeated value = 15

$$\text{Range} = 25 - 12 = 13$$

(max - min)

2. Construct a frequency distribution for

2, 3, 4, 2, 5, 3, 4, 2, 6, 5, 3, 4.

Value =	2	3	4	5	6	Total = 12
Frequency =	3	3	3	2	1	

2 occurs 3 times
3 occurs 3 times
4 occurs 3 times
5 occurs 2 times
6 occurs 1 time

3. Find Q_1 , Q_2 , Q_3 and IQR for:

5, 8, 10, 12, 15, 18, 20, 22, 23, 30

there are 10 values

$$Q_2 (\text{Median}) = \frac{15+18}{2} = 16.5$$

$$Q_2 = 16.5$$

Q1 lower half:

5, 8, 10, 12, 15

middle value (median) = 10

$$\therefore Q1 = 10$$

Q3 upper half:

18, 20, 22, 25, 30

middle value (median) = 22

$$\therefore Q3 = 22$$

IQR

$$IQR = Q3 - Q1$$

$$= 22 - 10 = 12$$

$$IQR = 12$$

4° Detect outliers using IQR method

For: 10, 12, 13, 15, 16, 17, 18, 20, 45.

using the median of values method:

$$Q1 = 12.5$$

$$Q3 = 19$$

$$IQR = Q3 - Q1 = 19 - 12.5 = 6.5$$

lower fence:

$$Q1 - 1.5(IQR)$$

$$= 12.5 - 1.5(6.5)$$

$$= 2.75$$

Upper fence :

$$Q3 + 1.5(IQR)$$

$$= 19 + 1.5(6.5)$$

$$= 28.75$$

Any value below 2.75 or above 28.75 is
an outlier

$$45 > 28.75$$

$\therefore$ outlier = 45

5. Determine the skewness (A symmetry) of

5, 6, 7, 7, 8, 8, 9, 20.

20 is larger than the other values

$$\text{mean} = \frac{70}{8} = 8.75$$

$$\text{median} = \frac{7+8}{2} = 7.5$$

since $\text{mean} > \text{median}$

the distribution is positively
skewed (right skewed).

6. A uniformly distributed variables range
from 10 to 80. Find $P(20 < X < 35)$

$$P(20 < X < 35)$$

For a uniform distribution:

$$P = \text{Required Interval} / \text{Total Interval}$$

Required Interval:

$$35 - 20 = 15$$

Total Interval:

$$50 - 10 = 40$$

Therefore:

$$P = 15/40$$

$$= 0.375$$

$$= 37.5\%$$

$$P(20 < x < 35) = 0.375 \text{ or } 37.5\%$$

7. A normal distribution has mean 100 and std deviation 15. Find the z-score for

$$x = 130$$

Given

$$\text{mean } (\mu) = 100$$

$$\text{std deviation } (\sigma) = 15$$

$$x = 130$$

$$\text{Formula: } z = \frac{(x - \mu)}{\sigma}$$

$$= \frac{130 - 100}{15} = \frac{30}{15} = 2$$

$$\boxed{z = 2}$$

8. For the same normal distribution
And $P(X < 130)$.

From Q7) $z = 2$

From the std normal table:

$$P(Z < 2) = 0.9772$$

$$\therefore P(X < 130) = 0.9772 = 97.72\%$$

9. Given a dataset with one extreme value, compare the mean & median before and after outlier treatment.

Consider the dataset:

10, 12, 13, 14, 15, 16, 17, 18, 100

Here 100 is an extreme value.
Before removing outlier 100

$$\text{mean} = \frac{215}{9} = 23.89$$

$$\text{median} = 15$$

After removing the outlier 100

dataset: 10, 12, 13, 14, 15, 16, 17, 18

$$\text{mean} = \frac{115}{8} = 14.375$$

$$\text{median} = \frac{(14+15)}{2} = 14.5$$

Measure	Before outlier	After outlier
Mean	23.89	14.375
Median	15	14.5

∴ Outlier greatly increases the mean, while the median changes only slightly

10. Perform a complete EDA on a small student-mark dataset and report central tendency, dispersion, distribution, outliers & skewness

Student marks:

45, 50, 52, 55, 58, 60, 62, 65, 68, 70, 75

A. Central tendency

Mean:

$$\text{Sum} = 680$$

$$\text{no. of students} = 11$$

$$\text{mean} = 680/11 = 61.82$$

Median:

$$\text{middle value} = 60$$

Mode:

no. value is repeated.

therefore no mode.

B. Dispersion

$$\text{Range} = \text{max} - \text{min} = 75 - 45 = 30$$

the marks have a range of 30 marks
std deviation is approximately 13.87 marks

C. Distribution:

Most marks are below 45 & 70 while
as 95 much higher than the other marks
therefore the distribution has a right tail

D. Outlier Distribution

sorted Data: 45, 50, 52, 55, 58, 60, 62, 65, 68, 70

$$Q1 = 52$$

$$Q3 = 68$$

$$IQR = 68 - 52 = 16$$

Upper fence

$$= Q3 + 1.5(IQR) = 68 + 1.5(16) = 92$$

Since: $95 > 92$

95 is an outlier

E. Skewness

Since the high value, 95, pulls the
mean upward: $\text{mean} = 61.82 > \text{median} = 60$

$\therefore$ the data is positively / right skewed